TEXORA · LECTURE SHEET

Shed and Shedding

The opening made in the warp sheet so that a pick can pass: how the warp divides into two layers, what decides the size and shape of that opening, and why a clear, well-timed shed is the condition for every other loom motion to work.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 3 · LEVEL 2 TERM 1

1. What a shed is — and why it must form

Weaving interlaces two sets of yarn at right angles: the **warp**, running lengthwise through the loom, and the **weft** (the **pick**), laid across it one pick at a time. For a pick to pass through the warp, the warp sheet cannot stay as one flat layer. It must be divided, for a short part of each loom cycle, into two separated layers — an upper layer and a lower layer — with a clear opening between them. That opening is the **shed**. The operation that creates it is **shedding**.

Once the shed is open, the pick is inserted through it — by shuttle on a conventional loom, or by projectile, rapier, air or water jet on modern machines (Topic 10). The reed beats the pick into the **cloth fell**, the shed changes — up layer down, down layer up — trapping the pick just laid. Repeating shed → pick → beat-up is weaving itself: with a bad shed there is no reliable weaving, because every later motion depends on the opening being clear, large enough and correctly timed.

Keep three ideas separate: the **shed** is the opening itself, **shedding** is the action of forming it, and the **shedding mechanism** — tappet, dobby or jacquard — is the system that decides which ends move. This sheet deals with the shed and its geometry; Topics 4–6 deal with the mechanisms.

2. The warp line and shed geometry

2.1 The warp line

The **warp line** is the path followed by a warp end through the loom when the shed is closed: from the warp beam at the back, over the back rest, through the drop wire and the eye of its heald, through a dent of the reed, and on to the cloth fell, from where the woven cloth passes over the breast beam to the cloth roller. When the shed is closed all ends lie, ideally, in one straight, level plane along this line. Shedding displaces ends above and below that plane; the further an end is displaced, and the longer it must stay displaced, the more it is stretched and abraded. Good shed design is therefore a compromise: open the warp only as much, and only as long, as picking actually needs.

2.2 The three dimensions of a shed

- **Shed length** — the distance along the warp from the **heald eye** to the **cloth fell**, the section free to open: behind the heald the ends are guided as one sheet, and at the fell the cloth fixes them together again. A longer length gives the same height with a smaller angle and less strain, but needs more heald movement for a given clearance at the shuttle.
- **Shed height (shed depth)** — the vertical distance between the layers at a stated position, most usefully where the shuttle passes. It is greatest at the heald and falls to zero at the fell, where both layers meet. Half the height — one layer's displacement from the closed line — is used in angle calculations for a symmetrical shed.
- **Shed angle** — the angle between one displaced warp layer and the closed warp line (or, when quoted as the full angle, the angle between the two layers). A small angle means gentle treatment of the warp; too small an angle at the race, however, leaves too little vertical clearance for the shuttle.

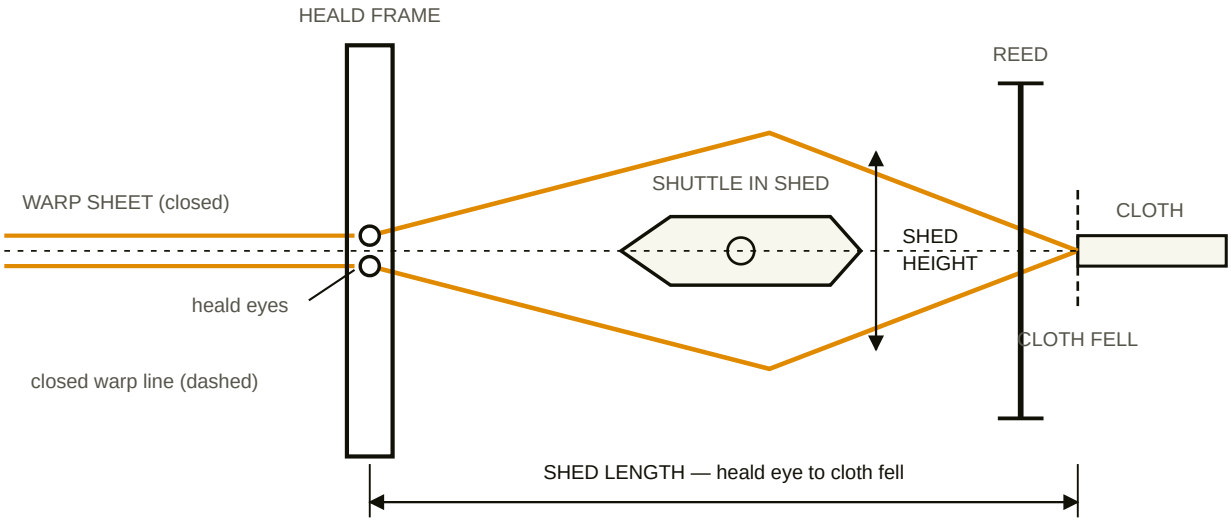


Figure 1 — Shed geometry in side view (cross-section along the warp). The warp sheet divides at the heald eyes into upper and lower layers, the shuttle passes through the opening, and both layers meet again at the cloth fell. Shed height therefore falls from a maximum at the healds to zero at the fell.

Because the two layers converge at the fell, shed height at any point follows from simple similar triangles. If an end is lifted by an amount H at the heald eye and the fell is a distance L in front of the eye line, then at a distance x from the eye toward the fell the lift remaining is $H \times (1 - x/L)$. This is why the shuttle race and the reed must be considered together with heald lift: the clearance available where the shuttle actually travels is always less than the lift imposed at the heald.

> **WORKED EXAMPLE — SHED ANGLE**
>
> A shed has a half shed height of 2.2 cm (one layer displaced 2.2 cm from the closed warp line) over a shed length of 25 cm. Find the shed angle of that layer to the warp line.

tan θ = opposite ÷ adjacent = 2.2 ÷ 25 = 0.088.

θ = $\tan^{-1}(0.088) \approx$ **5.0°**.

The full angle between the two symmetrically opened layers is therefore about 2 × 5.0° = **10.0°**. A small angle like this is gentle on the warp — the price paid for it is the long shed length needed to obtain enough height for the shuttle.

WORKED EXAMPLE — SHED HEIGHT FROM HEALD LIFT

A heald lifts its warp ends by 5.0 cm at the heald eye. The cloth fell is 30 cm in front of the eye line. What lift remains 20 cm from the eye toward the fell — for example, near the back of the shuttle race?

By similar triangles the lift falls linearly from 5.0 cm at the eye to 0 at the fell.

Height at 20 cm = 5.0 × (1 − 20 ÷ 30) = 5.0 × (10 ÷ 30) = 5.0 × 0.333… = **1.67 cm**.

So although the heald moves 5.0 cm, the warp at that point has risen only about 1.67 cm above the closed line. If the lower layer is lowered symmetrically, the total shed height there is about 2 × 1.67 = **3.33 cm** — the figure that must be checked against the shuttle height plus clearance.

3. Heald frames and heald eyes

Warp ends do not move individually on tappet and dobby looms. Each end is drawn through the eye of a flat steel (or, on older looms, wire or cord) **heald**, and the healds that must move together are mounted as a set between two horizontal shafts or rails to form a **heald frame**, also called a **harness** or **shaft**. Lifting or lowering the frame lifts or lowers every end drawn through its healds by exactly the same amount and at exactly the same time. The weave decides the grouping: in plain weave, alternate ends go on two frames; in a twill or satin, ends repeat in a pattern spread over more frames (the drawing-in plan).

The **heald eye** — the small opening near the middle of each heald — is the point from which shed geometry is measured. It must pass the end and its knots without chafing yet guide it positively; all eyes should sit on a common eye line when closed; and the vertical travel of the eye *is* the heald lift, setting shed height everywhere in front of the healds through the similar-triangle relation of Section 2. Worn or bent healds are a direct cause of warp breaks and a ragged shed at the selvedges.

Plain weave needs only 2 frames (often 4 for load sharing), a simple twill 3–4, and complex weaves up to the capacity of the mechanism — the first reason the choice in Section 7 matters.

4. Requirements of a good shed

- **A clear shed.** Every end assigned up must be fully up and every end assigned down fully down, with none floating in the middle — a floating end causes a shuttle trap, broken pick or stitch. Clarity depends on correct drawing-in, sound heald eyes, equal warp tension, and timing that lets the shed settle before picking.
- **Shuttle clearance.** Shed height at the race must exceed shuttle height (plus pirn and weft) by a safe margin throughout the passage — at entry and exit as well as mid-shed, where the shed is still opening or already closing.
- **Minimum warp strain.** Lifting an end lengthens its path from heald to fell, stretching it beyond its closed length. Excess lift, a short shed length or a very unsymmetrical shed raise tension peaks and end breakage. The target is the smallest lift giving safe clearance, shared as evenly as the design allows between the layers.
- **Correct timing and dwell.** Fully open before the shuttle enters, held open while it is inside, and not closing on its tail (Section 6).
- **No unnecessary movement.** Ends staying in the same layer for the next pick should stay there where the shed type allows — every avoidable reversal wastes energy and adds a tension cycle.

5. Types of shed by warp-sheet movement

Sheds are classified by what the two layers do *between* successive picks — whether they return to the closed line, meet in the middle, or move directly from one open position to the next.

TYPE	MOVEMENT BETWEEN PICKS	WARP STRAIN AND WEAR	NOTES AND TYPICAL USE
Bottom-closed shed	Both layers return to the bottom (closed) line after each pick, then reopen.	Highest: every end travels the full distance twice per pick.	Traditional shed of early looms. Simple, but wasteful of movement.
Centre-closed shed	Both layers return to the centre (warp) line after each pick, then reopen through the centre.	Moderate and balanced: each layer travels half the distance each way.	Common on tappet and dobby looms; the closed instant at the centre helps grip the pick just beaten up.
Open shed	Layers do *not* return to a closed line. Ends staying in the same layer stay put; only ends that must change move, crossing directly.	Lowest for repeats: an end staying in one layer for several picks does not move at all on those picks.	Standard where ends stay in one layer for several picks (dobby/jacquard). Saves movement and permits higher speeds.
Semi-open shed	Ends staying in the same layer remain; ends changing layer	Between centre-closed and open in movement	Used on some dobby looms to combine open-shed economy

	first return toward the closed line, then cross.	and crossing severity.	with a more positive change for moving ends.

The classification is about the *path* of the warp sheet, not the mechanism: tappet looms are normally associated with a closed shed, while open and semi-open sheds are characteristic of dobby and jacquard shedding. In every type the shed is judged by the same tests of Section 4.

6. Shed timing relative to the crank cycle

All primary loom motions are timed against the rotation of the **crankshaft**, which makes one revolution (360°) per pick. By convention 0° is taken at beat-up — the reed at its front position, pressing the new pick into the fell — and angles increase as the crank turns away from the fell. Shedding, picking and beat-up must fit inside this one circle without overlapping in a way that traps the shuttle.

Just after beat-up the shed begins to change and the new shed forms. Picking begins when the shed is open enough at the entering side — commonly around 90–120° of crank after beat-up on a conventional loom, though the exact figure is a loom setting. The shuttle crosses while the shed is held open (its **dwell** — Topic 4) and must leave before closing reduces exit-side clearance below shuttle height. Beat-up follows as the crank returns to 0°/360°.

If shedding is **too early**, the shed is already closing while the shuttle is inside — it is trapped or an end is broken. If **too late**, picking starts into a half-open shed; the shuttle rubs on the lower layer and may fly out or cut selvedge ends. Shed timing is therefore set together with picking timing, at the crank angles marked for the loom type.

7. Classification of shedding mechanisms — a preview

Three families of mechanism can produce the sheds described above. They differ not in the shed they aim at, but in how many ends they can control independently, how long a pattern repeat they can store, and at what cost and complexity.

MECHANISM	HOW THE PATTERN IS STORED	CONTROL AND CAPACITY	DEALT WITH IN
Tappet (cam) shedding	In the shape of rotating cams — one cam per heald frame	Each frame follows its cam. Simple weaves only; practically up to about 8–10 frames, short pick repeats.	Topic 4
Dobby shedding	In a pegged lattice, chain or electronic pattern read pick by pick	Each frame selected independently per pick. Medium patterns; commonly up to about 16–24 frames, longer repeats.	Topic 5

Jacquard shedding	In punched cards or electronic data, one record per pick	Ends (or small groups) controlled individually, not by frames. Large figured patterns; hundreds to thousands of hooks, repeats limited mainly by data length.	Topic 6

Tappet → dobby → jacquard is a progression in pattern freedom bought with complexity. The shed itself must satisfy the same requirements whichever mechanism forms it.

Glossary

Shed — the opening between the raised and lowered warp layers, through which the pick passes. **Shedding** — the motion that forms the shed. **Cloth fell** — the edge of the cloth where the newest pick is beaten in. **Warp line** — the straight warp path when the shed is closed. **Shed length** — heald eye to cloth fell. **Shed height (depth)** — the vertical distance between the two layers at a stated point. **Heald frame** — the frame carrying all healds that move together; also called a shaft. **Heald eye** — the opening in a heald through which one end is drawn. **Dwell** — the period the shed is held fully open without moving. **Crank angle** — crankshaft position in degrees within its 360° one-pick cycle, 0° at beat-up.

Key points

• A shed is the temporary division of the warp sheet into two layers; without it no pick can be inserted, and every other primary motion depends on it.
• Shed length runs from heald eye to cloth fell; shed height is greatest at the healds and falls linearly to zero at the fell; shed angle follows tan θ = half-height ÷ shed length.
• Heald lift at the eye overstates the clearance at the shuttle — use similar triangles to find the height where the shuttle actually travels.
• A good shed is clear, gives safe shuttle clearance, imposes the minimum warp strain, and is correctly timed with enough dwell.
• Bottom-closed, centre-closed, open and semi-open sheds differ in the path the layers take between picks, and therefore in movement, strain and suitability for repeated patterns.
• Tappet, dobby and jacquard are three ways of forming the same shed, in rising order of pattern capacity and complexity.

Practice

1. A shed has a half shed height of 1.8 cm over a shed length of 22 cm. Calculate the shed angle of one layer to the warp line, and the full angle between the layers.

2. A heald lifts its ends by 4.8 cm at the eye, and the cloth fell is 28 cm from the eye line. Calculate the lift remaining 14 cm from the eye toward the fell, and the total shed height there if the lower layer is lowered by the same amount.
3. Distinguish bottom-closed, centre-closed and open sheds by the path the warp layers take between two picks, and state which imposes the least movement on an end that stays in the same layer for four consecutive picks.
4. List the four requirements of a good shed and, for each, name one fault in the cloth or in weaving that follows when it is not met.
5. A loom traps the shuttle about once per hour, always near the exit side of the shed. Using shed timing, give two setting faults that could cause this and state what you would check first.
6. Explain why increasing heald lift is not a free cure for a tight shed: trace its effects on shed angle, warp strain and end breakage.

Answer key

1. $\tan\theta = 1.8 \div 22 = 0.0818$. $\theta = \tan^{-1}(0.0818) \approx$ **4.7°**. Full angle $\approx 2 \times 4.7° =$ **9.4°**.
2. Lift $= 4.8 \times (1 - 14 \div 28) = 4.8 \times 0.5 =$ **2.4 cm**. Total shed height with symmetrical lowering $= 2 \times 2.4 =$ **4.8 cm**.
3. Bottom-closed: both layers return to the bottom closed line after every pick, then reopen. Centre-closed: both layers return to the central warp line after every pick, then reopen. Open: layers do not return to a closed line; only ends that must change layer move. The **open shed** imposes least movement — an end staying in the same layer for four picks does not move at all on the intervening picks.
4. Any coherent set matching Section 4, for example: clear shed — otherwise shuttle traps and broken ends / stitches; shuttle clearance — otherwise shuttle rubs, cuts ends or flies out; minimum strain — otherwise end breakage rises and cloth becomes hairy; correct timing with dwell — otherwise the shed closes on the shuttle or picking starts into a half-open shed.
5. (i) Shedding too early, so the shed has begun to close at the exit side before the shuttle leaves; (ii) picking too late (or loom slowed unevenly) so the shuttle is still inside when closing starts. Check first: the crank angles of shed change and of picking against the loom's timing marks, and confirm the shed is still fully open at the exit side at the angle the shuttle should leave.
6. Greater lift also lengthens the path of every displaced end from heald to fell, so shed angle and warp extension rise with it. Tension peaks increase, ends are abraded more at eye and reed, and breakage rises. The cure is the smallest lift giving safe clearance, with shed length and timing checked first.
